

RAK14012 WisBlock LED Matrix Datasheet

Overview

Description

RAK14012 is an LED matrix driver module for WS2812B LEDs, which has a control circuit and RGB chip.

To drive the WS2812B, the RAK14012 has a built-in boost circuit to generate 5 V voltage output. But it is recommended to use an external power supply to power the LED Matrix.

Features

- LED matrix driver
- 3.3 V Power supply
- Module Size: 25 mm x 15 mm

Specifications

Overview

Mounting

The RAK14012 WisBlock LED Matrix can be mounted to the IO slot of the WisBlock Base board. **Figure 1** shows the mounting mechanism of the RAK14012 on a WisBlock Base board, such as the [RAK5005-O](#) .

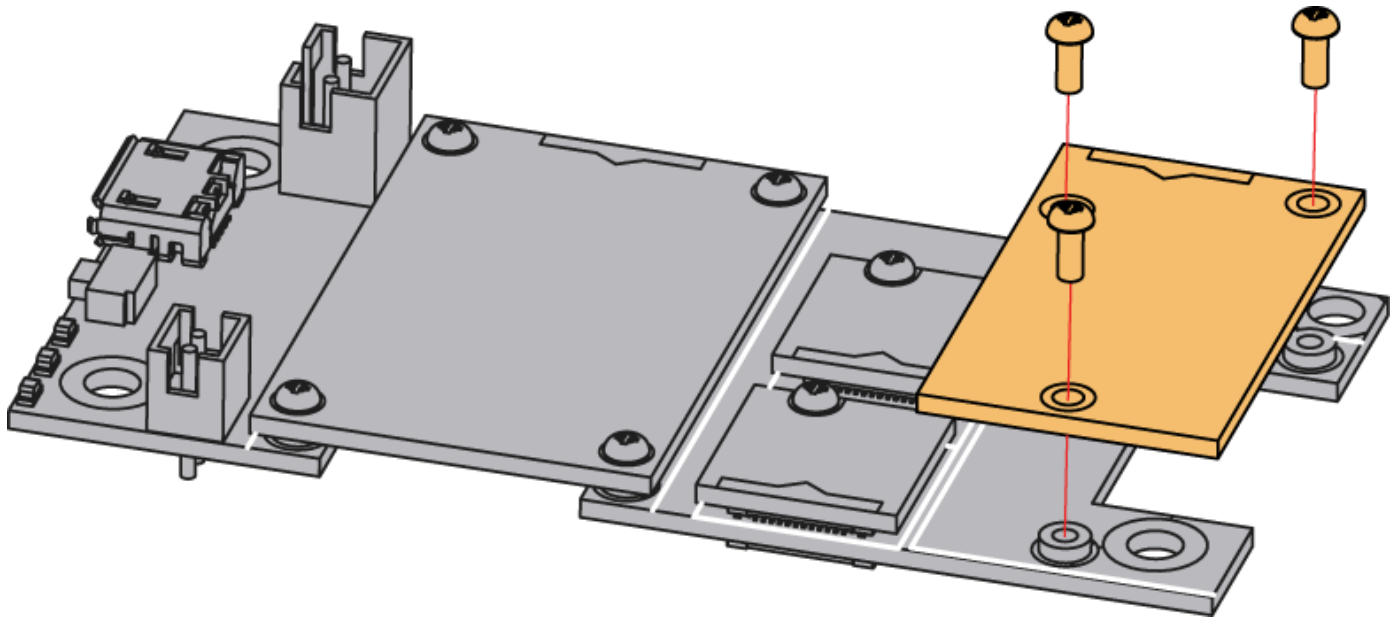


Figure 1: RAK14012 WisBlock LED Matrix mounting

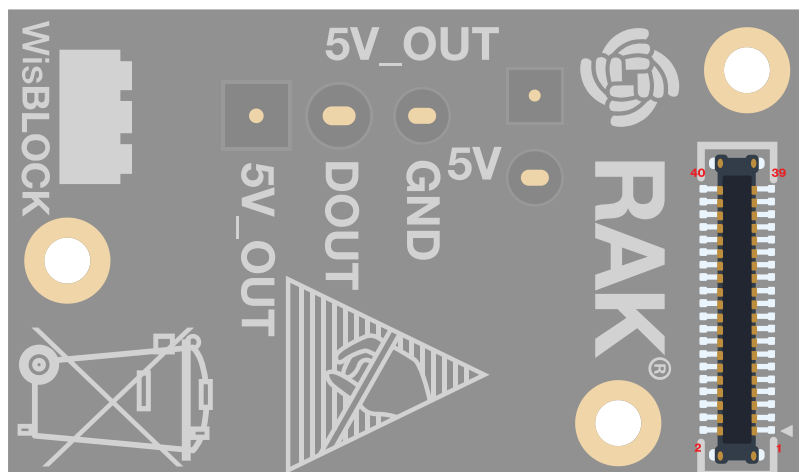
Hardware

Pin Definition

The RAK14012 WisBlock LED Matrix comprises a 40-pin WisConnector. The 40-pin WisConnector allows the RAK14012 module to be mounted to a WisBlock baseboard, such as RAK5005-O. The pin order of the connector and the pinout definition is shown in **Figure 2**.

NOTE

- The **VBAT**, **GND**, **EN (WB_IO6)**, and **DOUT (WB_IO5)** are connected to WisConnector.



VBAT	2	1	VBAT
GND	4	3	GND
3V3_S (Optional)	6	5	NC
NC	8	7	NC
NC	10	9	NC
NC	12	11	NC
NC	14	13	NC
NC	16	15	NC
NC	18	17	NC
NC	20	19	NC
NC	22	21	NC
NC	24	23	NC
NC	26	25	NC
DOUT (optional)	28	27	NC
NC	30	29	NC
NC	32	31	NC
NC	34	33	NC
NC	36	35	DOUT (optional)
EN (I06)	38	37	DOUT (I05)
GND	40	39	GND

Figure 2: RAK14012 WisBlock LED Matrix pinout

Electrical Characteristics

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
VBAT	Lithium battery output	Normal operation	3.3	-	4.2	V
3V3_S	3.3 V from WisBase	Normal operation	-	3.3	-	V
5V	Output from boost	Normal operation	-	5	-	V

Mechanical Characteristics

Board Dimensions

Figure 3 shows the dimensions and the mechanical drawing of the RAK14012 module.

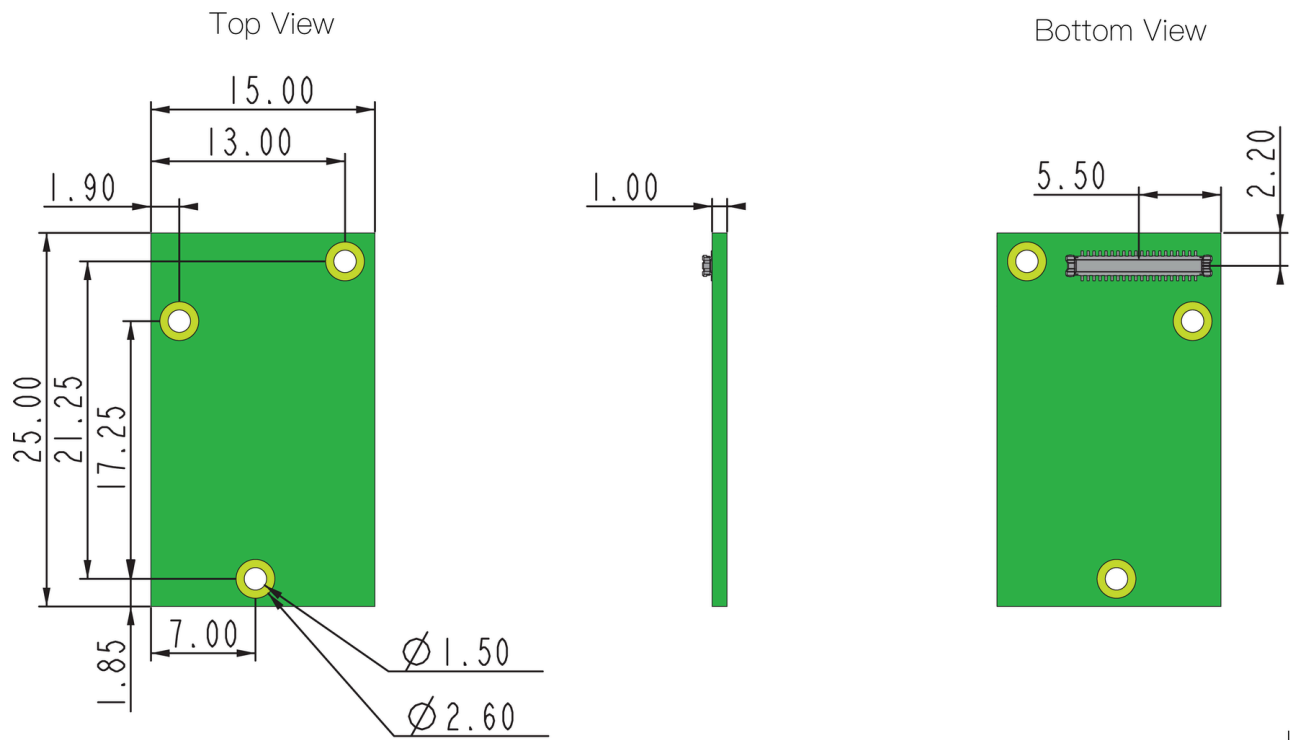


Figure 3: RAK14012 WisBlock LED Matrix dimensions

WisConnector PCB Layout

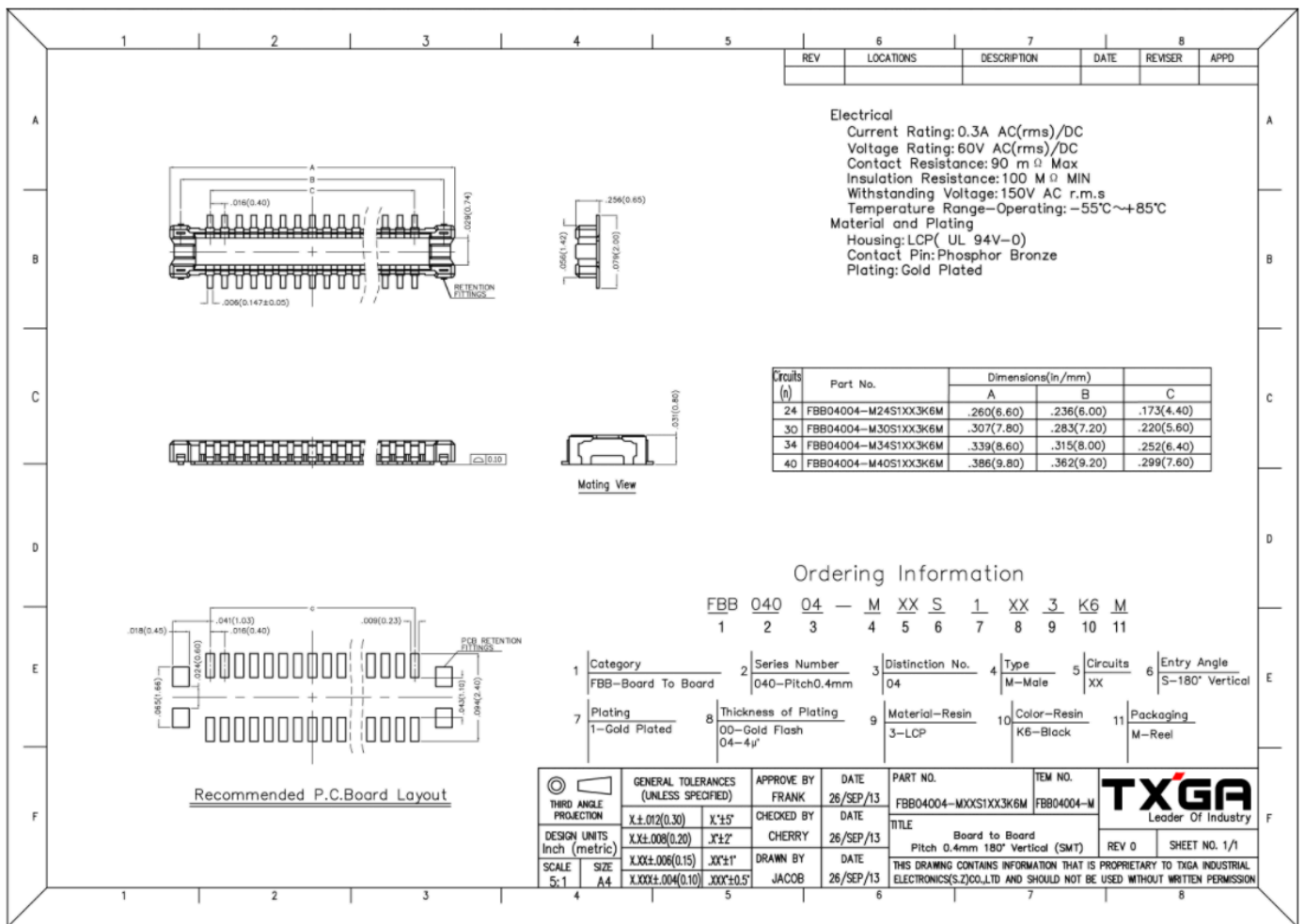


Figure 4: WisConnector PCB footprint and recommendations

Schematic Diagram

It has a boost circuit to generate 5 V because the supply for WS2812B must be between 3.7 V and 5.3 V. It is recommended to use an external power source to power the LED Matrix because of its high power consumption.

Connector

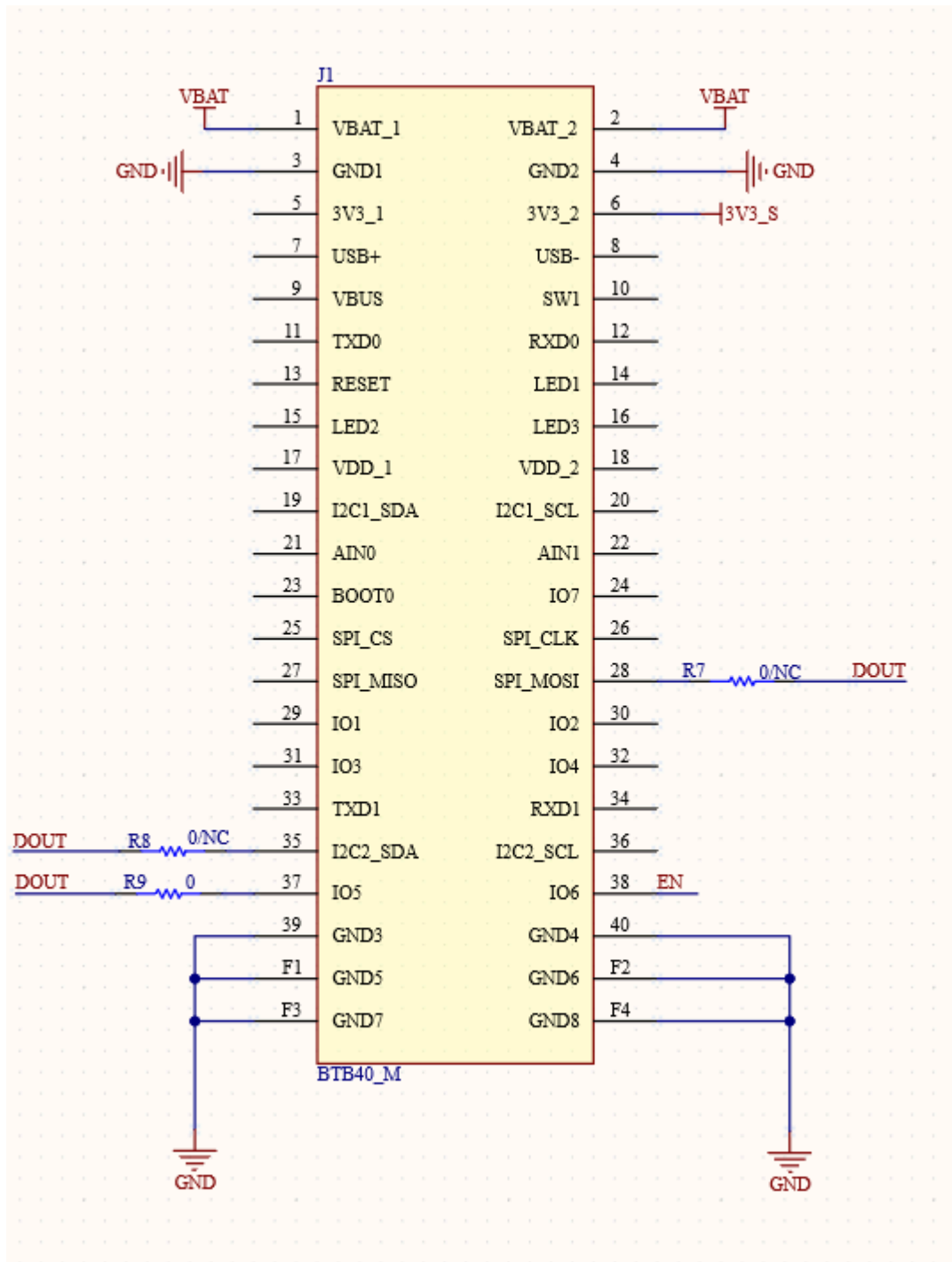


Figure 5: RAK14012 connector connection

Boost

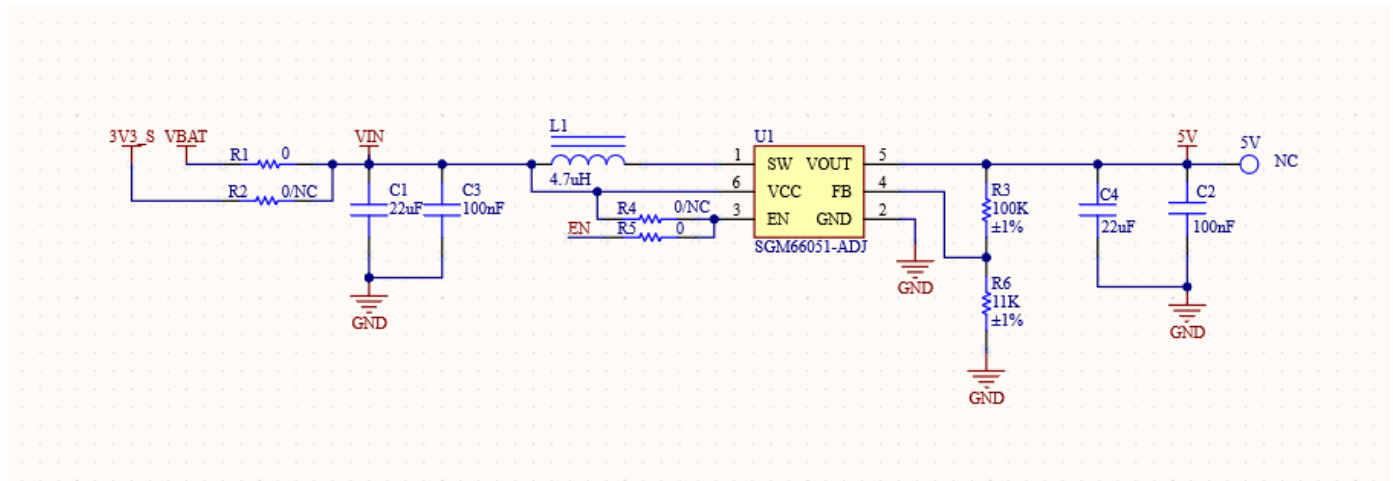


Figure 6: RAK14012 boost circuit

Power Selection

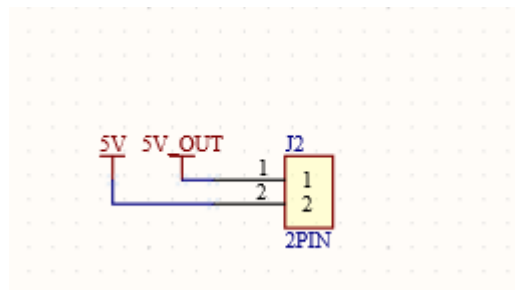


Figure 7: RAK14012 power selection

⚠ WARNING

- Make sure that the jumper from 5V_OUT of RAK14012 WisBlock LED Matrix is disconnected when connecting to an external power supply.
- LED Matrix DIN should be connected to RAK14012 DOUT.
- LED Matrix GND should be connected to RAK14012 GND.

Connector for LED Matrix

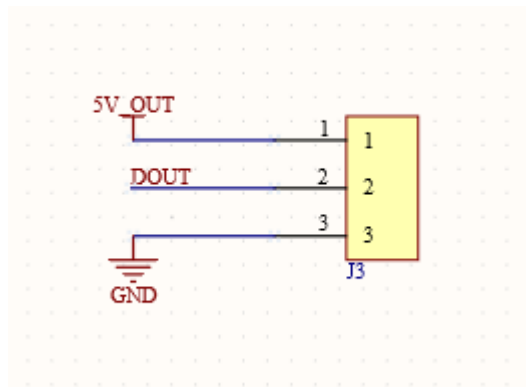


Figure 8: RAK14012 connector for LED matrix